

BERT’S SHEET METAL PRODUCTS

QUALITY MANUAL

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Prepared by:		Date:
Approved by:		Date:

Controlled document. The current revision is maintained in the BSMP shop management system; printed copies are uncontrolled.

1. Purpose and Scope

This manual defines the quality management system (QMS) of Bert's Sheet Metal Products ("BSMP"). It describes how quality is planned, controlled, documented, and verified for all products we quote, fabricate, inspect, and ship.

BSMP is not currently third-party registered to AS9100 or ISO 9001. This manual is written to align with the intent of those standards and serves as the governing quality document for the company. It applies to all employees and to every customer order.

2. Company Profile

Bert's Sheet Metal Products is a precision sheet-metal fabrication shop providing cutting, forming, punching, hardware insertion, welding, and assembly services. All work is performed at our facility.

Address / phone: _____

3. Quality System Overview — The BSMP Shop System

BSMP runs on a custom-built, cloud-based shop management system: a set of integrated applications sharing one real-time central database. Every record is stored centrally, timestamped, and attributed to the signed-in user. This system is the backbone of the QMS — procedures in this manual are implemented directly in these applications, so following the workflow is following the procedure.

- **Quote** — estimating and pricing, customer drawing storage, material and hardware price records.
- **Orders** — order entry, job tracking and routing, purchase-order pipeline, delivery and invoicing status, customer order confirmations.
- **QC** — inspections (start-of-job, operator self-checks, first articles, final), nonconformance reports (NCRs), material certifications library, drawing dimension checks, and a permanent quality audit log.
- **Mail** — controlled customer and vendor communication, RFQ and purchase-order capture, certification handoffs.
- **AP/AR** — vendor and receivables tracking, linked to purchase orders and vendor invoices.

Access is role-based: manager accounts have full access; operator accounts are limited to the Orders and QC applications. The QC application maintains an audit log recording every quality action (inspections, NCRs raised and closed, approvals) with user and timestamp.

4. Document and Data Control

Answers survey items 1, 2, 5, and 6.

- This Quality Manual is the top-level quality document. Quality procedures are defined in this manual and enforced by the workflows built into the shop system.
- Customer drawings and specifications are stored digitally and attached to the quote, the order, and the QC inspection records for the job. Because there is a single current copy in the central database,

every employee at every workstation sees the same, current revision — applicable documents are always available to the people responsible for product quality.

- Drawing revisions are verified at order entry against the customer purchase order; superseded revisions are replaced so shop personnel cannot work to an obsolete document.
- When work is subcontracted or material is purchased to a specification, the purchase order issued from the system carries the applicable drawings and requirements to the supplier.

5. Contract Review and Order Entry

- Quotes are prepared in the Quote application from the customer's drawing, including material, hardware, and process requirements.
- Incoming customer purchase orders are reviewed against the quote (part number, revision, quantity, price, due date) before acceptance.
- An order confirmation is emailed to the customer from the system, and the order record tracks the job through production, delivery, and invoicing.

6. Work Instructions

Answers survey items 3 and 4.

- Every job has an order record that serves as the job traveler: part number, revision, quantity, routing / process notes, due dates, and the linked customer drawing. It is prepared for each job at order entry.
- Work instructions include quality criteria: the controlling document is the customer drawing, and the QC application's Drawing Check extracts the drawing's dimensions and tolerances into the inspection checklist for the job.

Unless the customer drawing specifies otherwise, BSMP applies these standard title-block tolerances:

Dimension type	Standard tolerance
Fractional	$\pm 1/16''$
Two-place decimal (.XX)	$\pm .020''$
Three-place decimal (.XXX)	$\pm .005''$
Angular	$\pm 1^\circ$

7. Purchasing and Supplier Control

Answers survey items 6 and 10.

- BSMP maintains an approved vendor list in the shop system. Purchases of material, hardware, finishing, and outside processing are placed with approved vendors.

- Vendors are approved and retained based on their quality history and delivery performance, which is visible in the system: every purchase order, receipt, and vendor invoice is recorded and linked, so supplier problems are documented against the vendor.
- Purchase orders follow a controlled need → issue → receive pipeline in the Orders application. Requirements, specifications, and applicable drawings are transmitted with the PO.
- Received items are checked against the PO before the PO is closed as received.

8. Receiving and Raw Material Control

Answers survey item 9.

- Incoming material is verified at receipt against the purchase order for type, gauge/size, and quantity.
- Mill and material certifications are collected and stored in the QC application's Material Certs library, where they are linked to the vendor and to the jobs that consume the material, and are retrievable for customer review.
- Material in the shop is identified and traceable to its job through the order record; stock levels are tracked in the system's inventory records.

9. Customer-Furnished Material

Answers survey item 14.

- Customer-furnished material is inspected at receipt for condition, identity, and quantity.
- It is identified with the customer's order and stored so it cannot be mixed with other stock.
- Any loss, damage, or discrepancy is recorded and reported to the customer before further processing.

10. Inspection and Product Acceptance

Answers survey item 7.

Inspection is documented in the QC application at the following stages. Every inspection record is signed by the inspecting user and timestamped.

- **Receiving inspection** — purchased material and outside-processed parts are verified against the PO and drawing (Section 8).
- **Start-of-job / first-piece inspection** — a Start Inspection is opened with photos of the first part; the part is checked against the drawing, assisted by the system's automated drawing comparison, then routed to a self-check or first article.
- **In-process inspection** — operators record self-checks against the drawing's dimensions during the run. Flagging a failed self-check automatically raises an NCR (Section 15).

- **First article inspection** — formal first-article reports record actual measured values against each drawing dimension and tolerance. First articles are reviewed and approved by a manager. Out-of-tolerance dimensions automatically raise an NCR.
- **Final inspection** — parts are inspected before shipment; the order is not closed for delivery with an open NCR against it.

All automated inspection calculations show their work: the system displays the math and permanently retains the raw measured inputs alongside the result, so every acceptance decision can be independently verified.

11. Inspection Status

Answers survey items 8 and 19.

- The inspection status of every job is clear in the system: each inspection record and order carries status indicators (pass, flagged, open NCR, first-article approval state) visible to all users.
- BSMP does not use physical inspection stamps. Electronic acceptance records — user identity plus timestamp on each inspection sign-off, retained in the audit log — serve as the equivalent, with better traceability than a stamp.

12. Calibration

Answers survey items 11, 12, and 13.

- Measuring and test equipment used to verify or accept product (calipers, micrometers, height gauges, pin gauges, scales) is listed on the BSMP calibration register with an assigned ID and calibration due date.
- Instruments are calibrated at a defined interval (annually unless otherwise noted) by an accredited outside calibration service; calibration is traceable to N.I.S.T. and certificates are kept on file.
- Where checks are performed in-house between calibrations, they follow the instrument manufacturer's procedure against a calibrated reference, and are recorded.
- An instrument that is past due, damaged, or found out of tolerance is removed from service until recalibrated, and product accepted with it since the last good calibration is reviewed.

13. Quality Records

Answers survey item 15.

- Quality records — quotes, orders, inspections, first articles, NCRs, material certifications, purchase orders, and the audit log — are created and retained in the central database as work is performed.
- Records are retrievable by purchase order, part number, or date, and are maintained on file for customer review for a minimum of 7 years or as the customer's purchase order requires.
- The database is cloud-hosted with authenticated access; periodic backups are exported and retained.

14. Customer Right of Entry

Answers survey item 16.

Customers, their customers, and regulatory authorities have right of entry to BSMP's facility to review work, records, and processes applicable to their jobs, with reasonable notice during normal business hours.

15. Nonconformance and Corrective Action

Answers survey item 18.

- Nonconformances are documented as NCRs in the QC application. An NCR is raised by an operator flagging a part, automatically by an out-of-tolerance first-article dimension, or by a manager.
- Each NCR records the part, quantity affected, description of the nonconformance, and who raised it. Nonconforming product is segregated and identified until dispositioned.
- Dispositions are rework, scrap, or use-as-is (use-as-is only with customer approval). Corrective action taken is recorded on the NCR before it is closed, and closure includes review of the correction's effectiveness.
- All NCR activity is logged — raised, dispositioned, corrected, closed — with user and timestamp, so corrective actions are documented, logged, and reviewable for effectiveness and recurrence.

16. Internal Audit

Answers survey item 17.

- At least annually, management performs an internal audit of the QMS: a sample of completed orders is traced end-to-end (quote → order → inspections → shipment), open and closed NCRs are reviewed for trends, and the QC audit log is reviewed.
- Findings and resulting actions are documented and tracked to closure.

17. Training

Answers survey item 20.

- New employees are trained by the owner or a designated lead on the equipment, the shop system workflows, and the inspection requirements applicable to their role before working without supervision.
- System access enforces authorization by role: operator accounts are limited to the Orders and QC applications, and first-article approval is restricted to managers.
- Training is documented per employee (topics, trainer, date) and kept on file.

Appendix A — Supplier Quality Survey Cross-Reference

Index from the American Precision Tool & Engineering, Inc. Supplier Quality Survey (items 1–20) to this manual.

Item	Survey question	Manual section	Answer
1	Quality manual	This document	Yes
2	Quality procedures	Sections 3–17	Yes
3	Work instructions prepared for each job	Section 6	Yes
4	Work instructions include quality criteria	Section 6	Yes
5	Documents available to employees responsible for quality	Section 4	Yes
6	Documents supplied to suppliers	Sections 4, 7	Yes
7	Inspection documented (receiving, 1st piece, in-process, 1st article, final)	Section 10	Yes
8	Inspection status clear on all items	Section 11	Yes
9	Raw material controlled	Section 8	Yes
10	Suppliers controlled by survey or quality history	Section 7	Yes
11	Verification tools calibrated	Section 12	Yes
12	Calibration procedures for in-house calibration	Section 12	Yes
13	Calibrated tools traceable to N.I.S.T.	Section 12	Yes
14	Customer-furnished material inspected, stored, identified	Section 9	Yes
15	Quality records maintained for customer review	Section 13	Yes
16	Customer right of entry	Section 14	Yes
17	Internal audit within the past year	Section 16	Yes
18	Corrective actions documented, logged, reviewed	Section 15	Yes
19	Inspection stamps used	Section 11	No — electronic sign-off
20	Documented employee training	Section 17	Yes